SABIN BHANDARI

〒 761-0302 bhandari.sabin.nmc@gmail.com +81 80-8120-6817

香川県高松市上林町 507 レオパレスキュルティ上林 210 HTTPS://GITHUB.COM/SABINNMC

HTTPS://WWW.LINKEDIN.COM/IN/SABIN-BHANDARI-NMC/

SKILLS

- **Python (PyTorch)**
- Deep Neural Networks
- OpenCV (C++), Qt (C++)
- **Embedded Systems, Linux, I2C, CAN**
- Toradex Verdin iMX8M Plus (Torizon OS), Renesas R7FA6T1AD3CFP
- e^2 studio, LTspice, CSiEDA5
- **PWM, Microwave Analysis**
- **AutoCAD, IRONCAD 3D**

EDUCATION

Tribhuvan University, Bachelor of Engineering in Electrical Engineering

2014–2018

Nepal Engineering Council Certified Electrical Engineer

Professional Certification

HOBBY

Reading and finance

Travel

Artificial Intelligence

Financial-market research

Machine learning

LANGUAGE

English — Professional working proficiency

Japanese — JLPT N2

Hindi — Fluent

Nepali — Native

PROFILE

Embedded systems engineer with experience in computer vision and deep learning, supported by a strong foundation in C/C++ and Python. Hands-on experience with OpenCV, PyTorch, model fine-tuning, and embedded AI deployment. Comfortable working in multicultural environments, with JLPT N2 Japanese, professional working proficiency in English, native Nepali, and fluent Hindi. Passionate about solving complex engineering problems, continuously learning, and applying AI and open-source technologies to create accessible, practical solutions.

EXPERIENCE

Embedded System Engineer | PAL Giken Co., Ltd.

Jan 2024 – Present

Develop embedded systems and computer-vision solutions using Python and C/C++, including model fine-tuning and embedded AI deployment. Also contribute to electrical engineering and product design using IRONCAD 3D.

KEY CONTRIBUTIONS:

- Fine-tuned YOLOX for human detection, achieved 74% mAP, and developed a product using a Toradex Verdin iMX8M Plus SoM and Hailo AI processor.
- Fine-tuned RTMPose for real-time detection of eight container keypoints and combined it with YOLOX-based truck detection for large-truck parking safety.
- Analyzed data using principal component analysis (PCA) and visualized results with Seaborn.
- Developed a GUI-based prototype for controlling the virtual viewpoint of a fisheye camera.
- Implemented PWM signal analysis and control using a Renesas R7FA6T1AD3CFP microcontroller.
- Designed control-switch circuits and supported electrical design using LTspice and CSiEDA5.
- Developed I2C tools for reading, writing, and dumping data on a Toradex Verdin module running Torizon OS.

Educator | Computer Science & Programming | Cosmic International Academy

Feb 2021 – Dec 2023

Taught C/C++ programming to high-school and undergraduate students and guided them in web technologies including HTML, CSS, and JavaScript.

KEY CONTRIBUTIONS:

- Established a robotics club for students.

- Designed and delivered **HTML, CSS, and JavaScript** courses.
- Developed interactive curricula, emphasizing **practical coding skills** and real-world applications.

Project Engineer | ADMC Engineering Company

Nov 2018 – Dec 2020

Contributed to hydropower and transmission-line development, including 220 kV double-circuit transmission lines and high-voltage engineering.

KEY CONTRIBUTIONS:

- Conducted site surveys for the 220 kV double-circuit transmission line serving the 164 MW Kaligandaki Gorge Hydroelectric Project.
- Prepared technical reports for transmission-line licence applications submitted to Nepal's Department of Electricity Development (DoED).
- Led feasibility studies for the 132 kV double-circuit transmission line serving the Upper Khudi Hydropower Project.
- Surveyed and designed 11 kV and 33 kV high-voltage primary distribution lines across 18 cities.